



CL-2005 - Database Systems Lab

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Lab 1 - MS Access

Department of Software Engineering

National University of Computer and Emerging Sciences, Islamabad Campus

Lab 1: MS Access

Learning Objectives

- Understand the fundamental concepts of relational databases and their components
- Learn to create and manage database tables with appropriate data types and constraints
- Practice establishing relationships between tables to maintain data integrity
- Develop skills in using MS Access forms for efficient data entry and management
- Explore different form types and their appropriate use cases in database applications

Outcomes

- Students will be able to create and manipulate an MS Access database file
- Students will be able to design tables with primary keys, foreign keys, and appropriate field properties
- Students will demonstrate the ability to establish and maintain relationships between related tables
- Students will create multiple types of forms using the Form Wizard for effective data management
- Students will understand how to use forms for entering, viewing, and editing database records

Lab Tasks

CLO4: Develop and test database solutions using a relational database management system, including data definition and manipulation.

Task 1: Library Management System (10 marks)

In this task, you will create a database to manage a small library's collection and borrowing records. Imagine you are helping a school library organize information about their books and which students have borrowed them. Your database will consist of two main tables that are connected through a relationship.

Begin by creating a new MS Access database file. Once the database is created, you will build two tables from scratch using Design View. The first table should be named **Books** and will store information about the library's collection. This table needs the following fields with their appropriate data types:



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- **Book ID:** This should be the primary key and use the AutoNumber data type so that each book receives a unique identification number automatically
 - **Book Title:** Use Short Text to store the name of the book
 - **Author:** Use Short Text to store the author's name
 - **ISBN:** Use Short Text to store the International Standard Book Number
 - **Available Copies:** Use Number to track how many copies of this book are available for borrowing

The second table should be named **Borrowing_Records** and will track which students have borrowed which books. This table needs the following fields:

- **Record ID:** This should be the primary key with AutoNumber data type
- **Book ID:** This is a foreign key that will link to the books table, so use Number data type
- **Student Name:** Use Short Text to store the name of the student who borrowed the book
- **Borrow Date:** Use Date/Time to record when the book was borrowed
- **Return Date:** Use Date/Time to record when the book should be returned or was returned

After creating both tables in Design View, switch to Datasheet View and enter one sample record in each table to verify that your structure is correct. Your sample record **must have** a book titled "Introduction to Databases" with your name as the author, with an ISBN that starts with your roll number, and mark it as having three available copies. Then add a borrowing record for that same book showing that a student borrowed it on a recent date.

Next, establish a **one to many** relationship between the **books** table and the **borrowing_records** table using the **Book ID field** as the connecting key.

Finally, create the following forms using the Form Wizard:

1. A form that displays multiple borrowing records at once, making it easy to see all active loans in a list format
2. A split form for the books table that allows you to browse through all books while viewing detailed information about the selected book
3. A multiple items form for the Books table that displays several books at once in a list format, making it easy to browse through the entire collection

Also, add at least 4 new records using these forms.



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Task 2: Student Course Enrollment System (10 marks)

For this task, you will design a database that tracks students and the courses they are enrolled in. Think of this as a simplified version of FAST's registration system. You will create two related tables and use forms to manage enrollment information efficiently.

Begin by creating a new MS Access database file. Once the database is created, you will build two tables from scratch using Design View. The first table should be named **Students** and will store information about enrolled students. This table needs the following fields with their appropriate data types:

- **Student ID:** This should be the primary key and use the AutoNumber data type so that each student receives a unique identification number automatically
- **Student Name:** Use Short Text for the student's full name
- **Email:** Use Short Text for the student's email address
- **Major:** Use Short Text for the student's field of study
- **Enrollment Year:** Use Number to store the year the student enrolled

The second table should be named **Course_Enrollments** and will track which courses each student is enrolled in. This table needs the following fields:

- **Enrollment ID:** This should be the primary key with AutoNumber data type
- **Student ID:** This is a foreign key that will link to the Students table, so use Number data type
- **Course Code:** Use Short Text for the course identifier such as "CS101"
- **Course Name:** Use Short Text for the full name of the course
- **Credits:** Use Number for the credit hours the course is worth
- **Semester:** Use Short Text to indicate when the course is taken, such as "Fall 2024"

After creating both tables in Design View, switch to Datasheet View and enter one sample record in each table to verify that your structure is correct. Your sample record must have a student with your own name, your university email address, your actual major, and your enrollment year. Then add a course enrollment record for that same student showing they are enrolled in "CS2001 - Database Systems" worth 3 credits in the current semester. Next, establish a **one to many** relationship between the Students table and the Course_Enrollments table using the Student ID field as the connecting key.

Finally, create the following forms using the Form Wizard:



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1. A form that displays multiple course enrollment records at once, making it easy to see all course registrations in a list format
2. A split form for the Students table that allows you to browse through all students while viewing detailed information about the selected student
3. A multiple items form for the Course_Enrollments table that displays several course enrollments at once in a list format, making it easy to review all course registrations

Also, add at least 4 new records using these forms.

Task 3: Patient Information System (10 marks)

In this task, you will build a database for managing patient information and their appointments at a local medical clinic. This system will help keep track of who has appointments scheduled and when those appointments are taking place, to make it easier for the hospital management to keep track of their patients.

Begin by creating a new MS Access database file. Once the database is created, you will build two tables from scratch using Design View. The first table should be named **Patients** and will store information about the clinic's patients. This table needs the following fields with their appropriate data types:

- **Patient ID:** This should be the primary key and use the AutoNumber data type so that each patient receives a unique identification number automatically
- **Patient Name:** Use Short Text for the full name
- **Date of Birth:** Use Date/Time to store the patient's birth date
- **Phone Number:** Use Short Text for contact information
- **Blood Type:** Use Short Text for the patient's blood type

The second table should be named **Appointments** and will track scheduled appointments for each patient. This table needs the following fields:

- **Appointment ID:** This should be the primary key with AutoNumber data type
- **Patient ID:** This is a foreign key that will link to the Patients table, so use Number data type
- **Appointment Date:** Use Date/Time for when the appointment is scheduled
- **Appointment Time:** Use Short Text to store the time such as "10:30 AM"
- **Doctor Name:** Use Short Text for which doctor the patient will see
- **Purpose:** Use Short Text describing the reason for the visit



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After creating both tables in Design View, switch to Datasheet View and enter one sample record in each table to verify that your structure is correct. Your sample record must have a patient with your name, a date of birth that matches yours, a phone number that starts with your roll number, and your blood type. Then add an appointment record for that same patient showing they have an appointment scheduled for tomorrow with Dr. Zod for a "General Checkup". Next, establish a **one to many** relationship between the Patients table and the Appointments table using the Patient ID field as the connecting key.

Finally, create the following forms using the Form Wizard:

1. A form that displays multiple appointment records at once, making it easy to see all scheduled appointments in a list format
2. A split form for the Patients table that allows you to browse through all patients while viewing detailed information about the selected patient
3. A multiple items form for the Patients table that displays several patients at once in a list format, making it easy to browse through all registered patients

Also, add at least 4 new records using these forms.

Submission Guidelines:

- Submit **MS Access File(s)** containing the work for the task(s).
- If there are multiple files, mention the lab task number in each file name, for example: **C_i241234_Lab1_Task1**, **C_i241234_Lab1_Task2**.
- Name the submitted file as follows: SectionLetter_RollNumber_LabNumber, for example: **C_i241234_Lab1**.
- The unethical usage of Generative AI is strictly prohibited and will result in 0 marks.
- **In case a demo is conducted, there will be deductions based on your answers to the questions asked.**